

Linear Algebra I — Week 3

Span, Linear Independence & Setting Up Linear Systems

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Linear combination and span. A linear combination of $v_1, \dots, v_k \in V$ is $a_1v_1 + \dots + a_kv_k$ with $a_i \in \mathbb{F}$. $\text{span}\{v_1, \dots, v_k\}$ is the set of all of them; $\text{span} \emptyset = \{\mathbf{0}\}$. It is a subspace, and the smallest subspace containing $v_1, \dots, v_k$.

Setting up the system. To test $w \in \text{span}\{v_1, \dots, v_k\}$ in $\mathbb{F}^n$, solve $a_1v_1 + \dots + a_kv_k = w$. Coordinatewise this is n equations in k unknowns, with augmented matrix whose columns are $v_1, \dots, v_k \mid w$. Solution exists $\iff w \in \text{span}$.

Linear independence. $v_1, \dots, v_k$ are *independent* if $a_1v_1 + \dots + a_kv_k = \mathbf{0}$ forces $a_1 = \dots = a_k = 0$; otherwise *dependent*. Test: solve the homogeneous system; only $\mathbf{0} \Rightarrow$ independent, a nonzero solution $\Rightarrow$ dependent (and it gives a relation).

Useful facts. A set containing $\mathbf{0}$ is dependent. One vector is independent iff it is nonzero. Two vectors are independent iff neither is a scalar multiple of the other. $v_1, \dots, v_k$ dependent $\iff$ some $v_j \in \text{span}$ of the others. If $w \in \text{span}\{v_1, \dots, v_k\}$, adding w does not enlarge the span. Independence and span depend on the field: $\{(1, i), (i, -1)\}$ is dependent over $\mathbb{C}$ but independent over $\mathbb{R}$.

Counting. More homogeneous unknowns than equations forces a nonzero solution; so more than n vectors in $\mathbb{F}^n$ are dependent.

Worked Examples

Example 1 · Membership in a span

Is $(4, 7) \in \text{span}\{(1, 2), (3, 5)\}$ in $\mathbb{R}^2$?

Solution.

1. Solve $a(1, 2) + b(3, 5) = (4, 7)$: $a + 3b = 4$, $2a + 5b = 7$.
2. From the first, $a = 4 - 3b$. Substitute: $8 - 6b + 5b = 7$, so $b = 1$, $a = 1$.
3. Check: $(1, 2) + (3, 5) = (4, 7)$.

Answer: Yes: $(4, 7) = 1(1, 2) + 1(3, 5)$.

Example 2 · Membership (positive)

Is $(1, 2, 3) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$.
2. Match: $a = 1$, $b = 2$; third coordinate $a + b = 3$ ✓.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$.

Example 3 · Membership (negative)

Is $(1, 2, 4) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. As before the combination is $(a, b, a + b)$.
2. Match: $a = 1$, $b = 2$, but then $a + b = 3 \neq 4$.

Answer: No.

Example 4 · Dependence, finding a relation

Test $\{(1, 2, 1), (2, 1, 0), (1, -1, -1)\}$ for independence.

Solution.

1. Solve $a(1, 2, 1) + b(2, 1, 0) + c(1, -1, -1) = \mathbf{0}$: $a + 2b + c = 0$, $2a + b - c = 0$, $a - c = 0$.
2. Third: $c = a$. First: $2a + 2b = 0$, so $b = -a$. Second: $2a - a - a = 0$ ✓ (automatically).
3. So $(a, b, c) = (1, -1, 1)$ is a nontrivial solution. Check: $v_1 - v_2 + v_3 = (1 - 2 + 1, 2 - 1 - 1, 1 - 0 - 1) = \mathbf{0}$.

Answer: Linearly dependent: $v_1 - v_2 + v_3 = \mathbf{0}$.

Example 5 · Independence in P_2

Show $\{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution.

1. Suppose $a(1 + x) + b(x + x^2) + c(1 + x^2) = \mathbf{0}$. Compare coefficients.
2. Constant: $a + c = 0$. x : $a + b = 0$. x^2 : $b + c = 0$.
3. So $a = -c$ and $b = -a = c$; then $b + c = 2c = 0$, so $c = 0$, hence $a = b = 0$.

Answer: Linearly independent.

Example 6 · The field matters

Is $\{(1, i), (i, -1)\}$ linearly independent in $\mathbb{C}^2$ over $\mathbb{C}$? Over $\mathbb{R}$?

Solution.

1. Note $i \cdot (1, i) = (i, i^2) = (i, -1)$.
2. Over $\mathbb{C}$: $i v_1 - v_2 = \mathbf{0}$, a nontrivial relation, so dependent.
3. Over $\mathbb{R}$: $a(1, i) + b(i, -1) = \mathbf{0}$ with real a, b gives first coordinate $a + bi = 0$, so $a = b = 0$: independent.

Answer: Dependent over $\mathbb{C}$; independent over $\mathbb{R}$.

Example 7 · Proof: dependence means one vector is redundant

Prove: if $v_1, \dots, v_k$ are linearly dependent, then some v_j is a linear combination of the others.

Solution.

1. Take a nontrivial relation $a_1 v_1 + \dots + a_k v_k = \mathbf{0}$ and pick j with $a_j \neq 0$.
2. Isolate: $a_j v_j = -\sum_{i \neq j} a_i v_i$.
3. Divide by a_j (allowed because $a_j \neq 0$ in a field): $v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i$.

Answer: $v_j \in \text{span}\{v_i : i \neq j\}$.

Example 8 · Describing a span by an equation

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

Solution.

1. $x(1, 1, 0) + y(0, 1, 1) = (x, x + y, y)$.
2. Matching: $x = a$, $y = c$, and $b = x + y = a + c$.

Answer: $a - b + c = 0$

Practice Problems

Part A — Linear Combinations and Span

1.

[Foundational]

Write $(5, -1)$ as a linear combination of $(1, 1)$ and $(1, -1)$ in $\mathbb{R}^2$.

2.

[Foundational]

Decide whether $(2, 3, 4) \in \text{span}\{(1, 1, 1), (0, 1, 2)\}$ in $\mathbb{R}^3$. If so, give the coefficients.

3.

[Foundational]

Decide whether $(1, 0, 0) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

4.

[Intermediate]

Find k such that $(1, 2, k) \in \text{span}\{(1, 1, 0), (1, -1, 2)\}$.

5.

[Intermediate]

In $\mathbb{C}^2$ over $\mathbb{C}$, write $(1 + i, 2)$ as a linear combination of $(1, i)$ and $(1, -i)$.

6.

[Foundational]

Show that $\text{span}\{(1, 2), (2, 4)\} \subseteq \mathbb{R}^2$ is a line, and give its equation.

7.

[Intermediate]

Describe $\text{span}\{(1, 0, 2), (0, 1, -1)\} \subseteq \mathbb{R}^3$ by a single linear equation in x, y, z .

8.

[Challenge]

Prove that for any vectors $v_1, \dots, v_k$ in a vector space V , the set $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Part B — Linear Independence and Dependence

9.

[Foundational]

Is $\{(1, 2), (3, 6)\}$ linearly independent in $\mathbb{R}^2$?

10.

[Foundational]

Is $\{(1, 2), (2, 3)\}$ linearly independent in $\mathbb{R}^2$?

11.

[Intermediate]

Is $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$ linearly independent in $\mathbb{R}^3$?

12.

[Intermediate]

Is $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ linearly independent in $\mathbb{R}^3$? If not, give a nontrivial relation.

13.

[Challenge]

Find all real k for which $\{(1, 1, 1), (1, 2, 3), (1, 3, k)\}$ is linearly dependent.

14.

[Foundational]

In $P_2(\mathbb{R})$, is $\{1 + x, 1 - x, 2\}$ linearly independent?

15.

[Intermediate]

In $\mathbb{C}^2$, consider $\{(1, i), (1+i, i-1)\}$. (a) Is it linearly independent over $\mathbb{C}$? (b) Is it linearly independent over $\mathbb{R}$?

16.

[Challenge]

Let u, v, w be linearly independent in a real vector space. Prove that $u + v, v + w, u + w$ are linearly independent.

Part C — Setting Up Systems, Spanning Sets and Proofs

17.

[Foundational]

Set up and solve the system that decides whether $(1, 2, 3) \in \text{span}\{(1, 0, 1), (2, 1, 0), (0, 1, 1)\}$.

18.

[Intermediate]

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 2, 0), (0, 1, 1)\}$.

19.

[Intermediate]

Find a finite spanning set for $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

20.

[Intermediate]

(a) Show that $\{(1, 1), (1, 2), (2, 1)\}$ spans $\mathbb{R}^2$. (b) Is it linearly independent?

21.

[Intermediate]

Show that $\{1, 1 + x, 1 + x + x^2\}$ spans $P_2(\mathbb{R})$.

22.

[Challenge]

Show that $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

23.

[Challenge]

Prove: if $w \in \text{span}\{v_1, \dots, v_k\}$ then $\text{span}\{v_1, \dots, v_k, w\} = \text{span}\{v_1, \dots, v_k\}$.

24.

[Challenge]

Let $v_1, \dots, v_k$ be linearly independent and $v_{k+1} \notin \text{span}\{v_1, \dots, v_k\}$. Prove that $v_1, \dots, v_{k+1}$ are linearly independent.